

# Report Generated by Test Manager

**Title:** Test Reports of SoC  
**Author:** Victoria  
**Date:** 10-Apr-2026 10:02:02

## Test Environment

Platform: PCWIN64  
MATLAB: (R2025b)

## Summary

| Name                                                                                                                    | Outcome | Duration<br>(Seconds) |
|-------------------------------------------------------------------------------------------------------------------------|---------|-----------------------|
| <a href="#">Results: 2026-Apr-10 09:09:46</a>                                                                           | ✓       | 14.515                |
|  <a href="#">New Test Case 1</a>       | ✓       | 7.178                 |
| <a href="#">Results: 2026-Apr-10 09:50:20</a>                                                                           | ✓       | 3.36                  |
|  <a href="#">worst case test</a>       | ✓       | 2.81                  |
| <a href="#">Results: 2026-Apr-10 09:56:41</a>                                                                           | ✓       | 1.982                 |
|  <a href="#">battery voltage test</a>  | ✓       | 1.717                 |
| <a href="#">Results: 2026-Apr-10 09:58:09</a>                                                                           | ✓       | 1.621                 |
|  <a href="#">cell temperature test</a> | ✓       | 1.33                  |
| <a href="#">Results: 2026-Apr-10 09:59:21</a>                                                                           | ✓       | 1.724                 |
|  <a href="#">low battery test</a>      | ✓       | 1.532                 |

## Results: 2026-Apr-10 09:09:46

Result Type: Result Set  
Parent: None  
Start Time: 10-Apr-2026 09:09:48  
End Time: 10-Apr-2026 09:10:03  
Outcome: Total: 1, Passed: 1

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## New Test Case 1

### Test Result Information

Result Type: Test Case Result  
Parent: [Results: 2026-Apr-10 09:09:46](#)  
Start Time: 10-Apr-2026 09:09:48  
End Time: 10-Apr-2026 09:09:55  
Outcome: Passed

### Test Case Information

Name: New Test Case 1  
Type: Simulation Test

### Logical and Temporal Assessments

| Name                                                                                            | Assessment                                                                                                                    |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|  Assessment1 | At any point in time, if (battery_current == 0) becomes true then, with a delay of at most 1 seconds, (soc == 0) must be true |

## Simulation

### System Under Test Information

Model: SOC\_Regler  
Release: Current  
Simulation Mode: normal  
Override SIL or PIL Mode: 0  
Configuration Set: Configuration  
External Input Name: battery\_current\_0\_input.mat

External Input File: C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04\_Verification\battery\_current\_0\_input.mat

Start Time: 0

Stop Time: 10

Checksum: 3846692330 2512201270 814111440 1202636617

Simulink Version: 25.2

Model Version: 1.4

Model Author: 320236370

Date: Wed Apr 08 19:47:39 2026

User ID: 320236370

Model Path: C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC\_Regler.slx

Machine Name: YY309528

Solver Name: ode45

Solver Type: Variable-Step

Max Step Size: 0.20000000000000001

Simulation Start Time: 2026-04-10 09:09:50

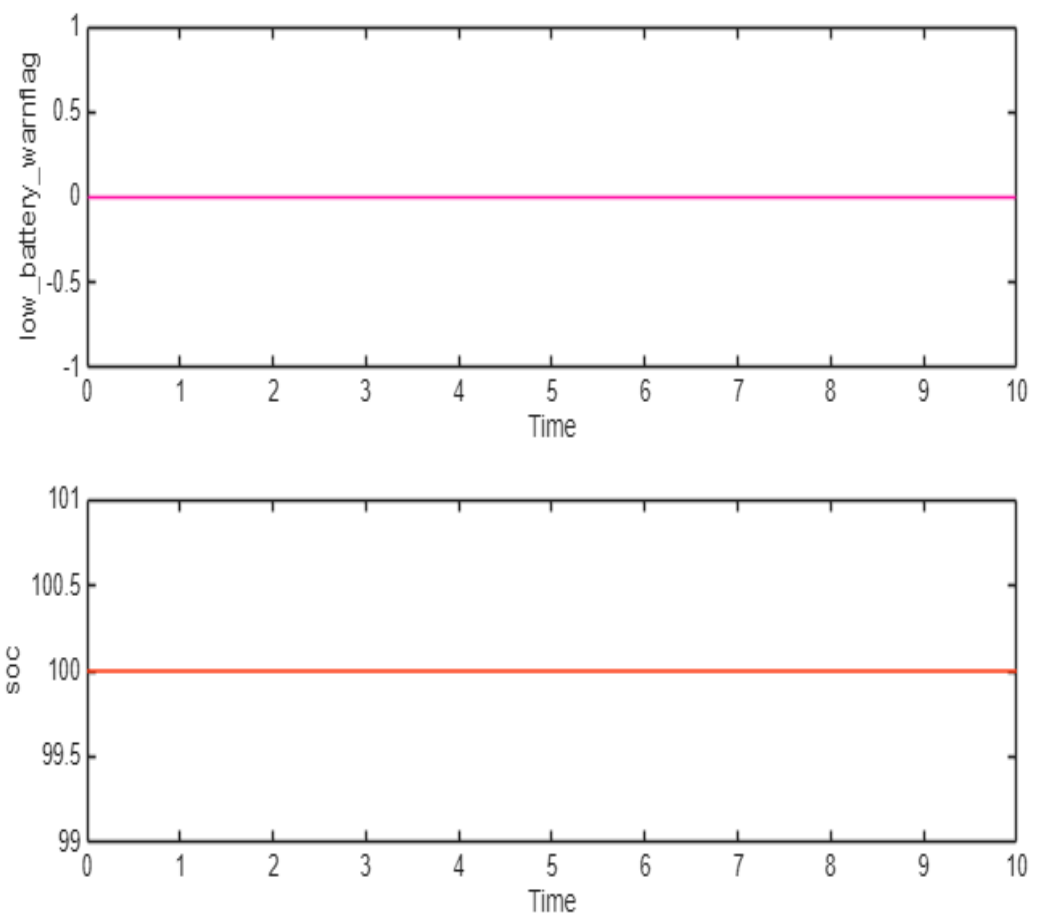
Simulation Stop Time: 2026-04-10 09:09:52

Platform: PCWIN64

## Simulation Output

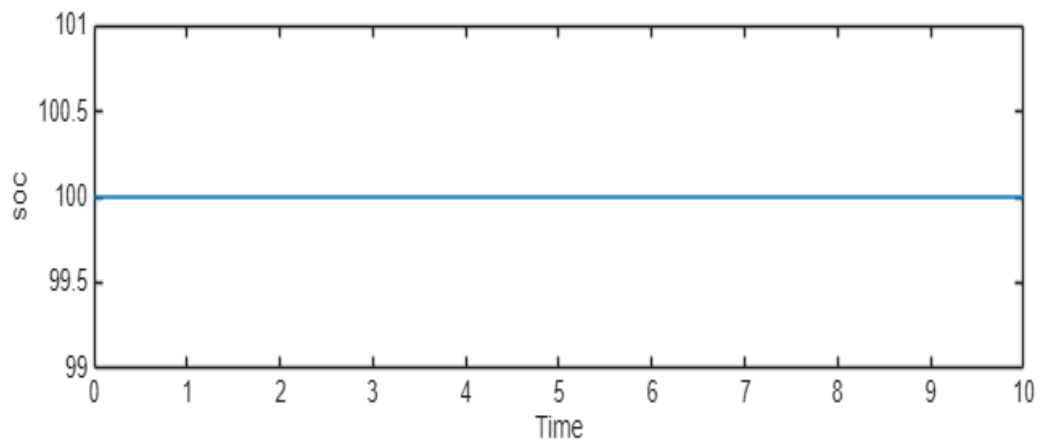
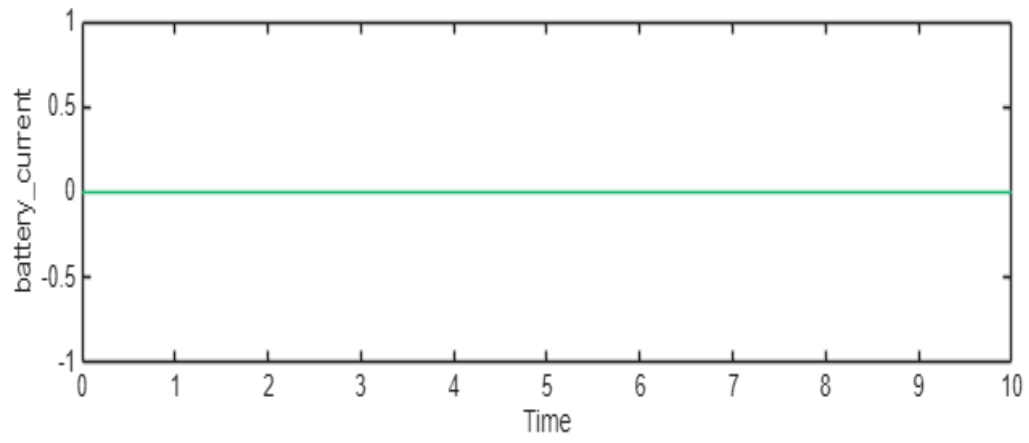
| Name                | Data Type | Units | Sample Time | Interp | Sync  | Link to Plot         |
|---------------------|-----------|-------|-------------|--------|-------|----------------------|
| low_battery_warning | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| soc                 | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| battery_current     | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| soc                 | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag         | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |

| Name                 | Data Type | Units | Sample Time | Interp | Sync  |
|----------------------|-----------|-------|-------------|--------|-------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union |
| soc                  | double    |       | Continuous  | linear | union |



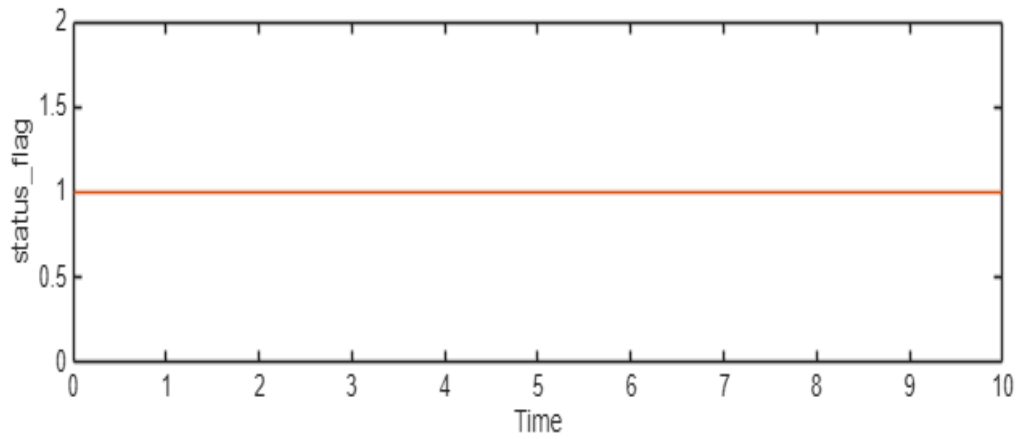
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| Name            | Data Type | Units | Sample Time | Interp | Sync  |
|-----------------|-----------|-------|-------------|--------|-------|
| battery_current | double    |       | Continuous  | linear | union |
| soc             | double    |       | Continuous  | linear | union |



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| Name        | Data Type | Units | Sample Time | Interp | Sync  |
|-------------|-----------|-------|-------------|--------|-------|
| status_flag | boolean   |       | Continuous  | zoh    | union |



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#### Simulation Logs:

Inputs may not be compatible for simulation. Test results might not be accurate. Click [here](#) for more information on external input mapping.

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## Results: 2026-Apr-10 09:50:20

Result Type: Result Set  
Parent: None  
Start Time: 10-Apr-2026 09:50:21  
End Time: 10-Apr-2026 09:50:24  
Outcome: Total: 1, **Passed: 1**

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## worst\_case\_test

### Test Result Information

Result Type: Test Case Result  
Parent: [Results: 2026-Apr-10 09:50:20](#)  
Start Time: 10-Apr-2026 09:50:21  
End Time: 10-Apr-2026 09:50:24  
Outcome: **Passed**

### Test Case Information

Name: worst\_case\_test  
Type: Simulation Test

### Logical and Temporal Assessments

| Name                                                                                            | Assessment                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Assessment1 | At any point in time, if (((cell_temperature > 90) & (battery_voltage <= 30)) & (soc < 20)) becomes true then, with a delay of at most 1 seconds, ((status_flag == true) & (low_battery_warningflag == true)) must be true |

## Simulation

### System Under Test Information

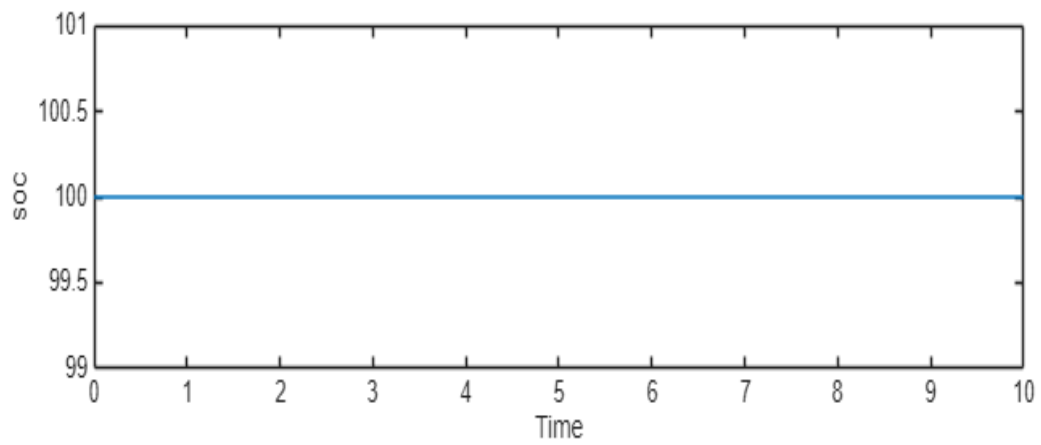
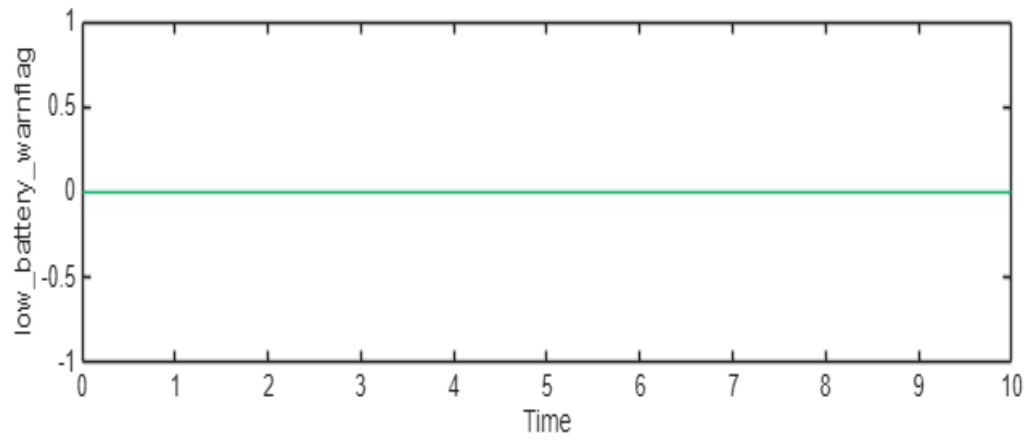
Model: SOC\_Regler  
Release: Current

|                           |                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Simulation Mode:          | normal                                                                                                                            |
| Override SIL or PIL Mode: | 0                                                                                                                                 |
| Configuration Set:        | Configuration                                                                                                                     |
| External Input Name:      | worst_case_input.mat                                                                                                              |
| External Input File:      | C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04_Verification\worst_case_input.mat |
| Start Time:               | 0                                                                                                                                 |
| Stop Time:                | 10                                                                                                                                |
| Checksum:                 | 3233302630 925315875 422518952 2889595808                                                                                         |
| Simulink Version:         | 25.2                                                                                                                              |
| Model Version:            | 1.4                                                                                                                               |
| Model Author:             | 320236370                                                                                                                         |
| Date:                     | Wed Apr 08 19:47:39 2026                                                                                                          |
| User ID:                  | 320236370                                                                                                                         |
| Model Path:               | C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC_Regler.slx                       |
| Machine Name:             | YY309528                                                                                                                          |
| Solver Name:              | ode45                                                                                                                             |
| Solver Type:              | Variable-Step                                                                                                                     |
| Max Step Size:            | 0.20000000000000001                                                                                                               |
| Simulation Start Time:    | 2026-04-10 09:50:21                                                                                                               |
| Simulation Stop Time:     | 2026-04-10 09:50:22                                                                                                               |
| Platform:                 | PCWIN64                                                                                                                           |

## Simulation Output

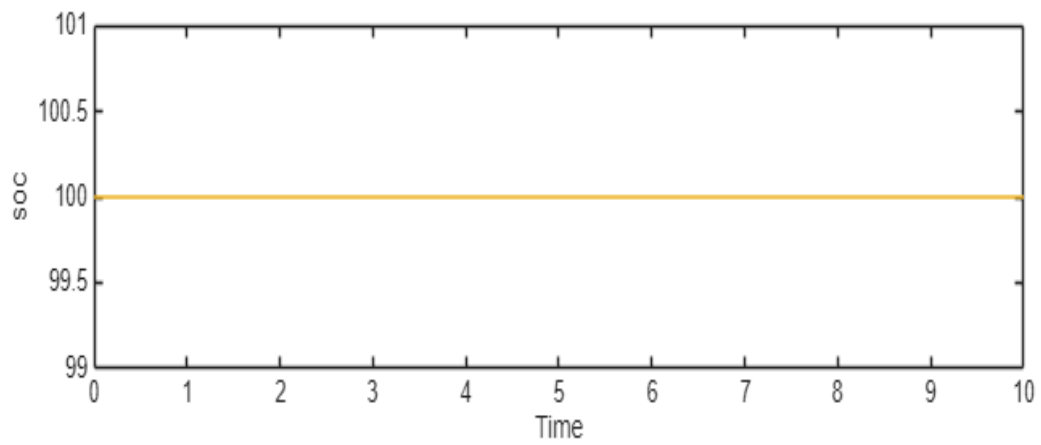
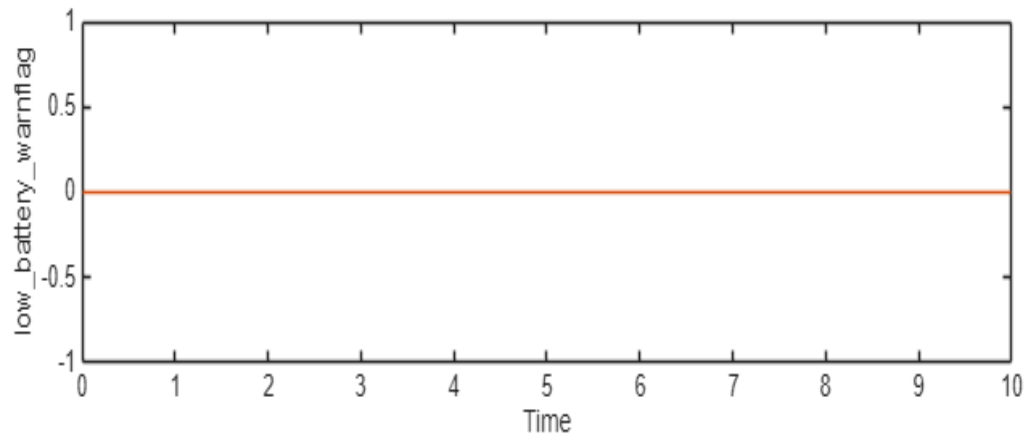
| Name                 | Data Type | Units | Sample Time | Interp | Sync  | Link to Plot         |
|----------------------|-----------|-------|-------------|--------|-------|----------------------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| soc                  | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| soc                  | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag          | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| battery_voltage      | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| cell_temperature     | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag          | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |

| Name                 | Data Type | Units | Sample Time | Interp | Sync  |
|----------------------|-----------|-------|-------------|--------|-------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union |
| soc                  | double    |       | Continuous  | linear | union |



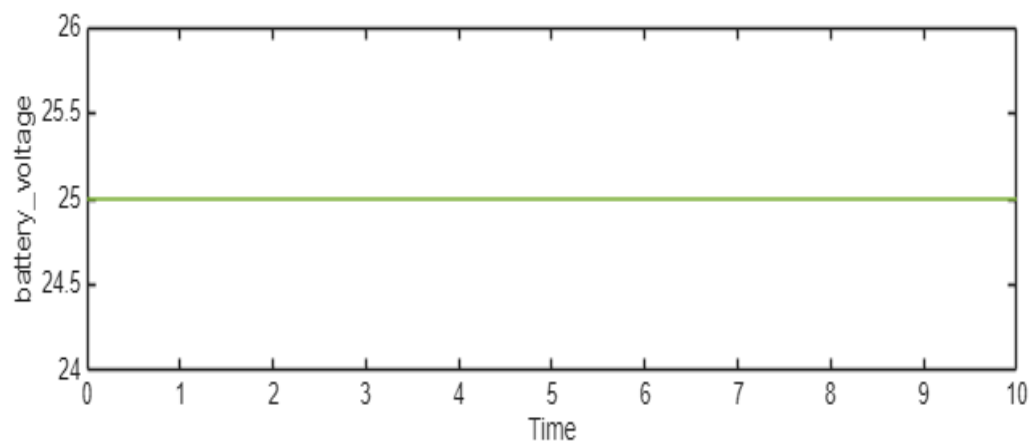
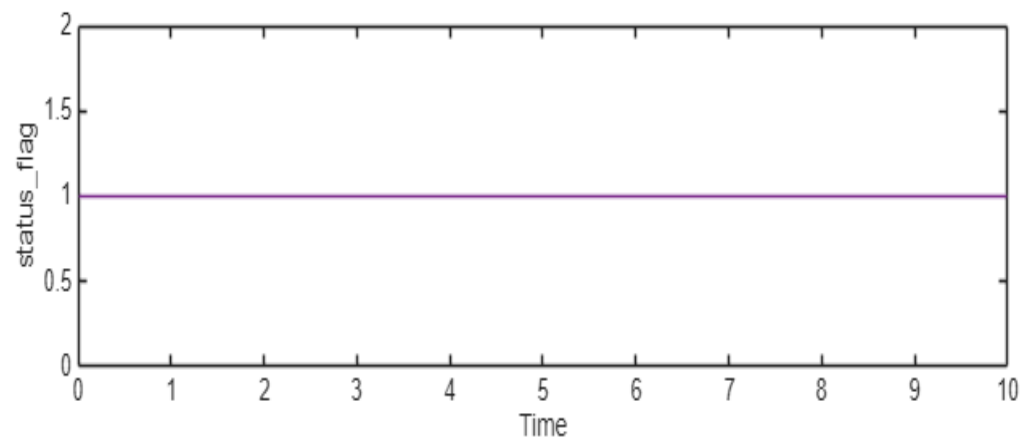
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| Name                 | Data Type | Units | Sample Time | Interp | Sync  |
|----------------------|-----------|-------|-------------|--------|-------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union |
| soc                  | double    |       | Continuous  | linear | union |



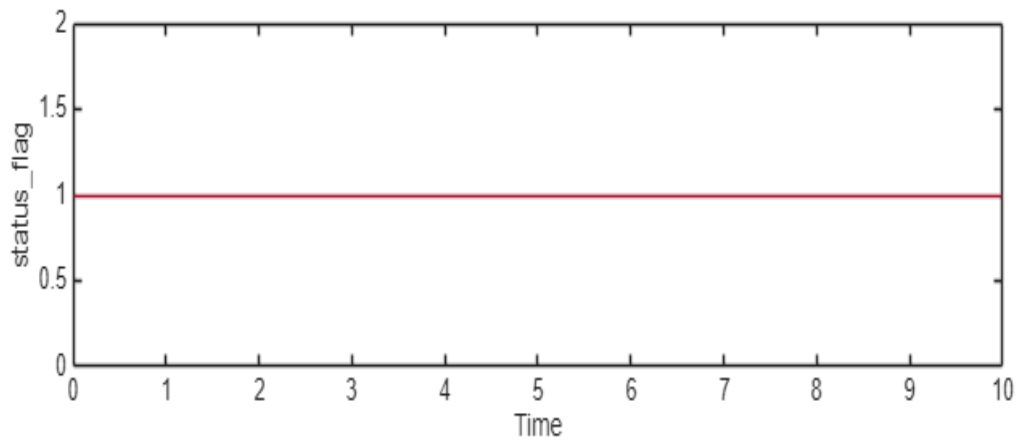
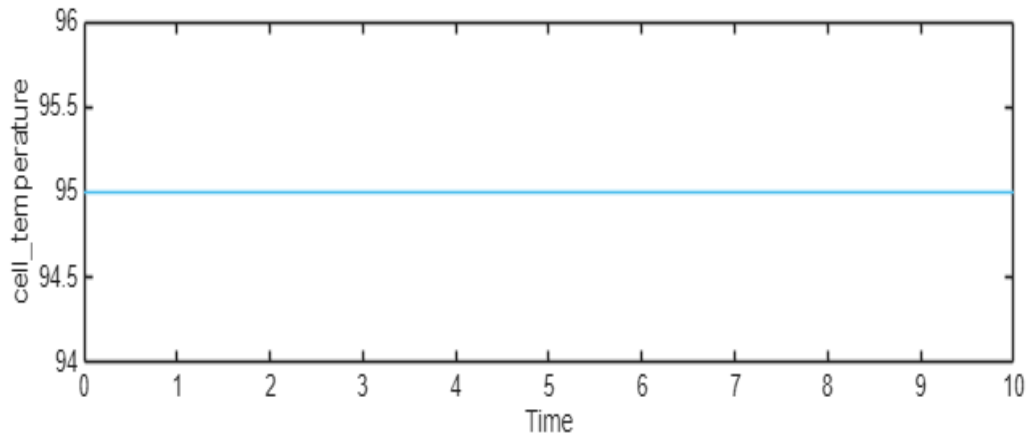
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| Name            | Data Type | Units | Sample Time | Interp | Sync  |
|-----------------|-----------|-------|-------------|--------|-------|
| status_flag     | boolean   |       | Continuous  | zoh    | union |
| battery_voltage | double    |       | Continuous  | linear | union |



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| Name             | Data Type | Units | Sample Time | Interp | Sync  |
|------------------|-----------|-------|-------------|--------|-------|
| cell_temperature | double    |       | Continuous  | linear | union |
| status_flag      | boolean   |       | Continuous  | zoh    | union |



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#### Simulation Logs:

Inputs may not be compatible for simulation. Test results might not be accurate. Click [here](#) for more information on external input mapping.

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## Results: 2026-Apr-10 09:56:41

Result Type: Result Set  
Parent: None  
Start Time: 10-Apr-2026 09:56:41  
End Time: 10-Apr-2026 09:56:43  
Outcome: Total: 1, **Passed: 1**

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## battery\_voltage\_test

### Test Result Information

Result Type: Test Case Result  
Parent: [Results: 2026-Apr-10 09:56:41](#)  
Start Time: 10-Apr-2026 09:56:41  
End Time: 10-Apr-2026 09:56:43  
Outcome: **Passed**

### Test Case Information

Name: battery\_voltage\_test  
Type: Simulation Test

### Logical and Temporal Assessments

| Name                                                                                            | Assessment                                                                                                                                |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|  Assessment1 | At any point in time, if (battery_voltage <= 30) becomes true then, with a delay of at most 1 seconds, (status_flag == true) must be true |

## Simulation

### System Under Test Information

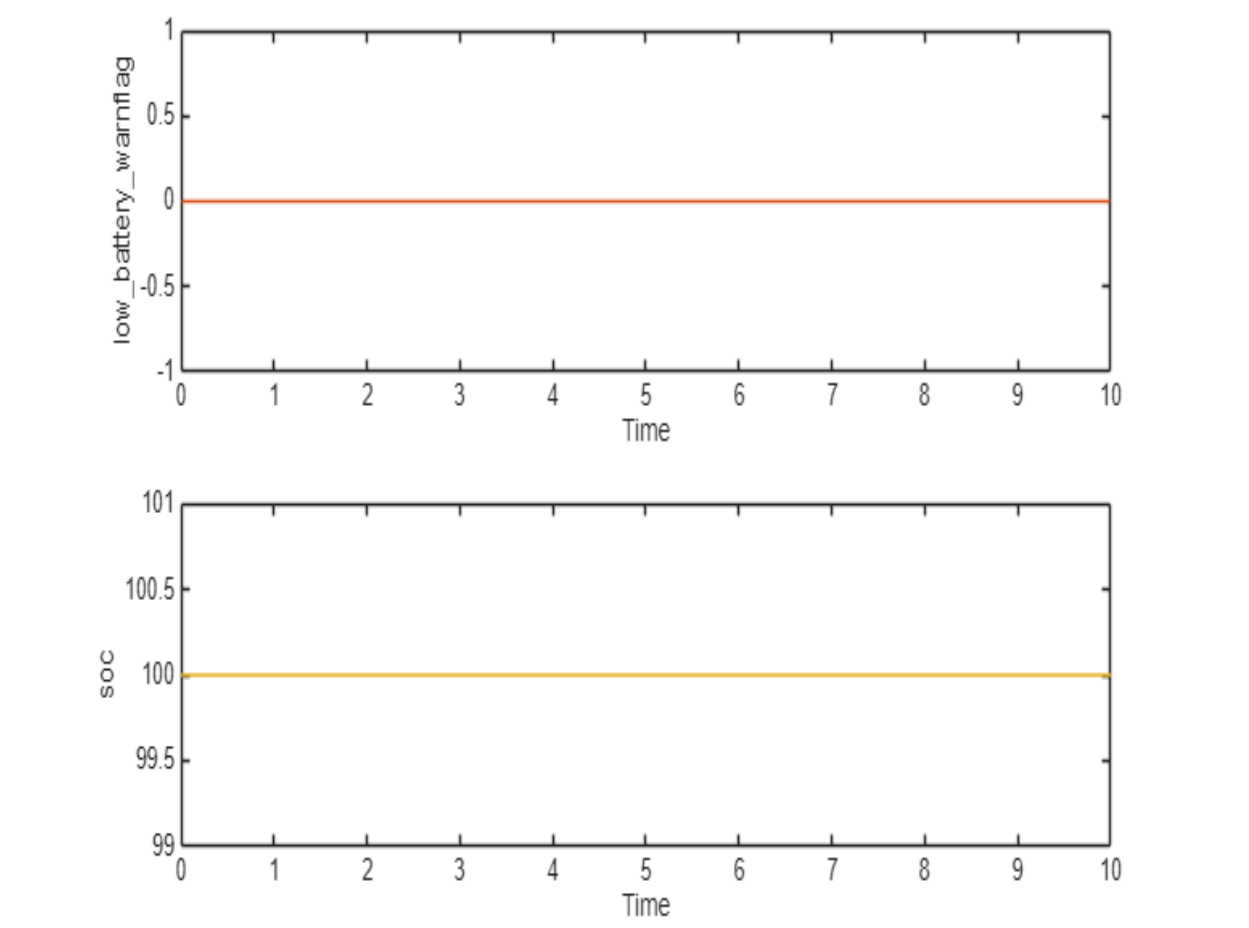
Model: SOC\_Regler  
Release: Current  
Simulation Mode: normal

Override SIL or PIL Mode: 0  
 Configuration Set: Configuration  
 External Input Name: battery\_voltage\_input.mat (1)  
 External Input File: C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04\_Verification\input\_data\battery\_voltage\_input.mat  
 Start Time: 0  
 Stop Time: 10  
 Checksum: 4132136549 3765386925 1652960231 359931920  
 Simulink Version: 25.2  
 Model Version: 1.4  
 Model Author: 320236370  
 Date: Wed Apr 08 19:47:39 2026  
 User ID: 320236370  
 Model Path: C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC\_Regler.slx  
 Machine Name: YY309528  
 Solver Name: ode45  
 Solver Type: Variable-Step  
 Max Step Size: 0.20000000000000001  
 Simulation Start Time: 2026-04-10 09:56:41  
 Simulation Stop Time: 2026-04-10 09:56:42  
 Platform: PCWIN64

## Simulation Output

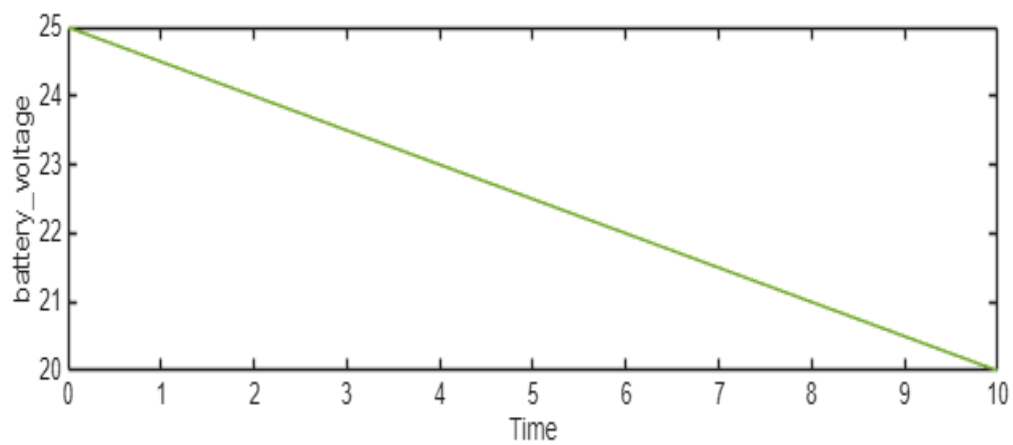
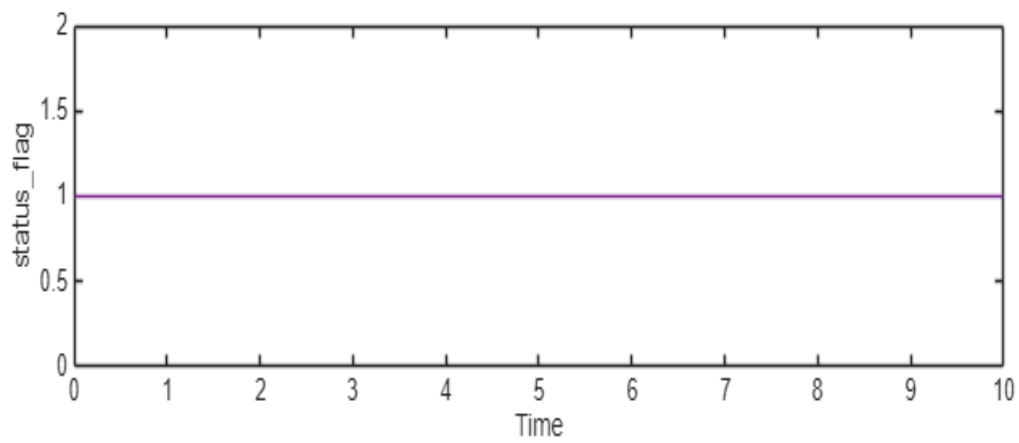
| Name                | Data Type | Units | Sample Time | Interp | Sync  | Link to Plot         |
|---------------------|-----------|-------|-------------|--------|-------|----------------------|
| low_battery_warning | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| soc                 | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag         | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| battery_voltage     | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag         | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |

| Name                 | Data Type | Units | Sample Time | Interp | Sync  |
|----------------------|-----------|-------|-------------|--------|-------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union |
| soc                  | double    |       | Continuous  | linear | union |



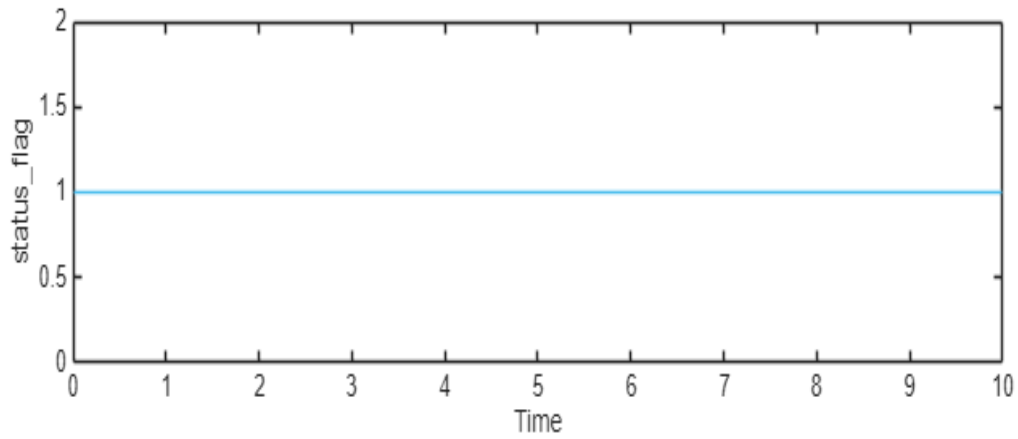
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| Name            | Data Type | Units | Sample Time | Interp | Sync  |
|-----------------|-----------|-------|-------------|--------|-------|
| status_flag     | boolean   |       | Continuous  | zoh    | union |
| battery_voltage | double    |       | Continuous  | linear | union |



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| Name        | Data Type | Units | Sample Time | Interp | Sync  |
|-------------|-----------|-------|-------------|--------|-------|
| status_flag | boolean   |       | Continuous  | zoh    | union |



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#### Simulation Logs:

Inputs may not be compatible for simulation. Test results might not be accurate. Click [here](#) for more information on external input mapping.

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## Results: 2026-Apr-10 09:58:09

Result Type: Result Set  
Parent: None  
Start Time: 10-Apr-2026 09:58:09  
End Time: 10-Apr-2026 09:58:10  
Outcome: Total: 1, **Passed: 1**

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## cell\_temperature\_test

### Test Result Information

Result Type: Test Case Result  
Parent: [Results: 2026-Apr-10 09:58:09](#)  
Start Time: 10-Apr-2026 09:58:09  
End Time: 10-Apr-2026 09:58:10  
Outcome: **Passed**

### Test Case Information

Name: cell\_temperature\_test  
Type: Baseline Test

### Logical and Temporal Assessments

| Name                                                                                            | Assessment                                                                                                                                |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|  Assessment1 | At any point in time, if (cell_temperature > 90) becomes true then, with a delay of at most 1 seconds, (status_flag == true) must be true |

## Simulation

### System Under Test Information

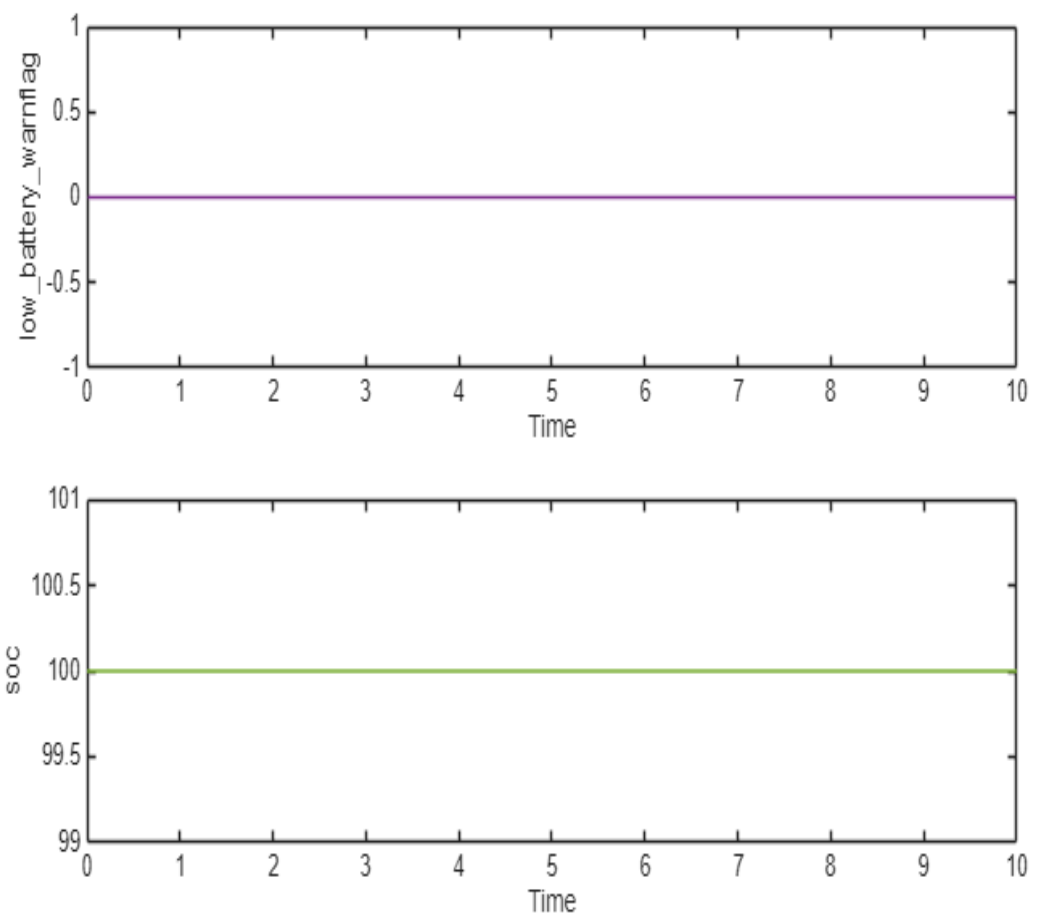
Model: SOC\_Regler  
Release: Current  
Simulation Mode: normal

Override SIL or PIL Mode: 0  
 Configuration Set: Configuration  
 External Input Name: cell\_temperature\_input.mat (1)  
 External Input File: C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04\_Verification\input\_data\cell\_temperature\_input.mat  
 Start Time: 0  
 Stop Time: 10  
 Checksum: 1396786863 1825109906 3322312545 4255407849  
 Simulink Version: 25.2  
 Model Version: 1.4  
 Model Author: 320236370  
 Date: Wed Apr 08 19:47:39 2026  
 User ID: 320236370  
 Model Path: C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC\_Regler.slx  
 Machine Name: YY309528  
 Solver Name: ode45  
 Solver Type: Variable-Step  
 Max Step Size: 0.20000000000000001  
 Simulation Start Time: 2026-04-10 09:58:09  
 Simulation Stop Time: 2026-04-10 09:58:09  
 Platform: PCWIN64

## Simulation Output

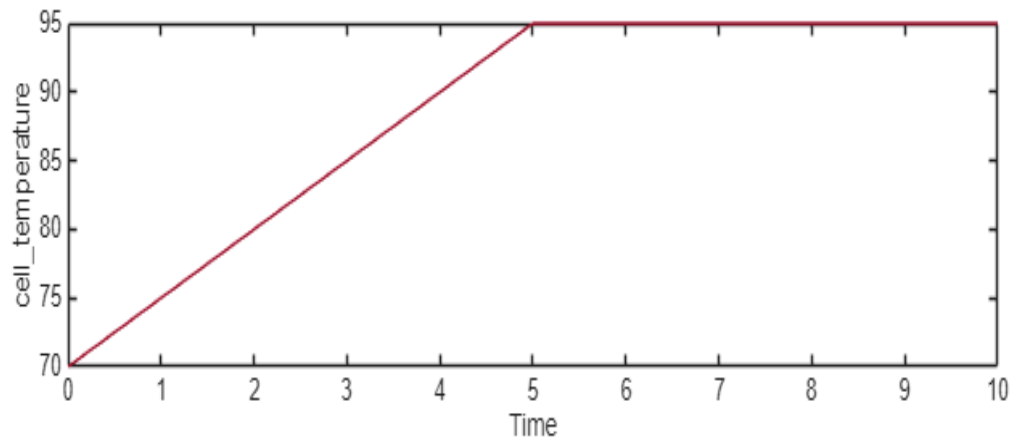
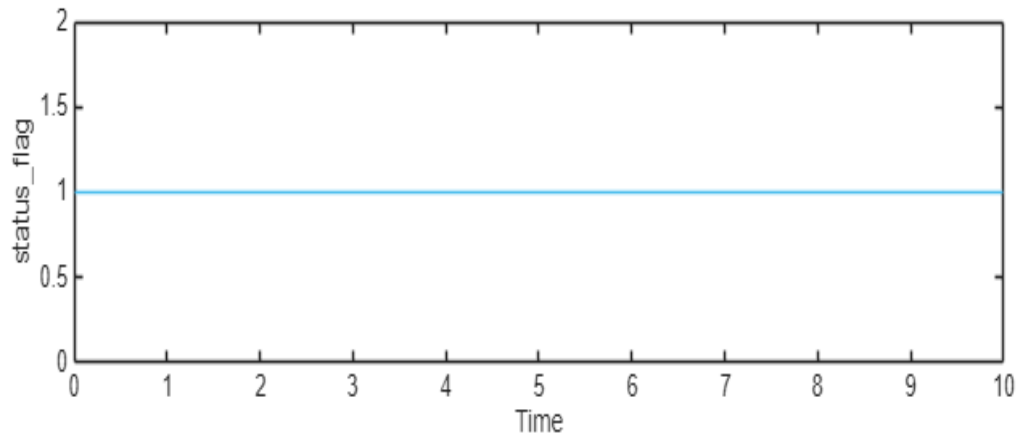
| Name                | Data Type | Units | Sample Time | Interp | Sync  | Link to Plot         |
|---------------------|-----------|-------|-------------|--------|-------|----------------------|
| low_battery_warning | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| soc                 | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag         | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| cell_temperature    | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag         | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |

| Name                 | Data Type | Units | Sample Time | Interp | Sync  |
|----------------------|-----------|-------|-------------|--------|-------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union |
| soc                  | double    |       | Continuous  | linear | union |



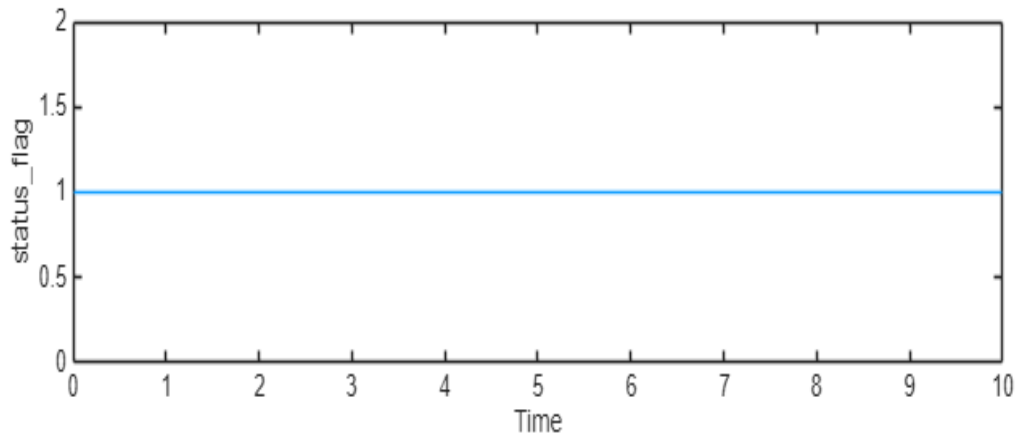
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| Name             | Data Type | Units | Sample Time | Interp | Sync  |
|------------------|-----------|-------|-------------|--------|-------|
| status_flag      | boolean   |       | Continuous  | zoh    | union |
| cell_temperature | double    |       | Continuous  | linear | union |



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| Name        | Data Type | Units | Sample Time | Interp | Sync  |
|-------------|-----------|-------|-------------|--------|-------|
| status_flag | boolean   |       | Continuous  | zoh    | union |



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#### Simulation Logs:

Inputs may not be compatible for simulation. Test results might not be accurate. Click [here](#) for more information on external input mapping.

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### Test Logs:

No baseline criteria evaluation performed as no baseline data is available for this test.

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## Results: 2026-Apr-10 09:59:21

Result Type: Result Set  
Parent: None  
Start Time: 10-Apr-2026 09:59:21  
End Time: 10-Apr-2026 09:59:22  
Outcome: Total: 1, **Passed: 1**

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## low\_battery\_test

### Test Result Information

Result Type: Test Case Result  
Parent: [Results: 2026-Apr-10 09:59:21](#)  
Start Time: 10-Apr-2026 09:59:21  
End Time: 10-Apr-2026 09:59:22  
Outcome: **Passed**

### Test Case Information

Name: low\_battery\_test  
Type: Simulation Test

### Logical and Temporal Assessments

| Name                                                                                            | Assessment                                                                                                                                       |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|  Assessment1 | At any point in time, if (state_of_current < 20) becomes true then, with a delay of at most 1 seconds, (low_battery_warnflag > 0.5) must be true |

## Simulation

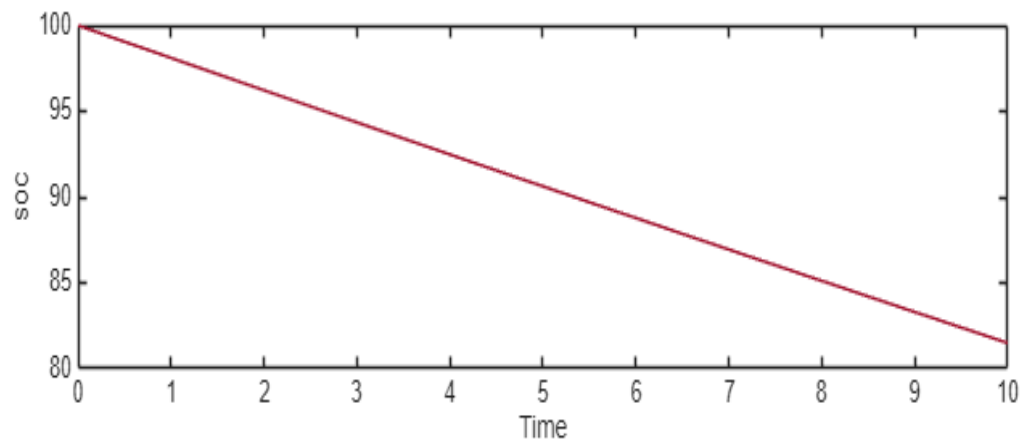
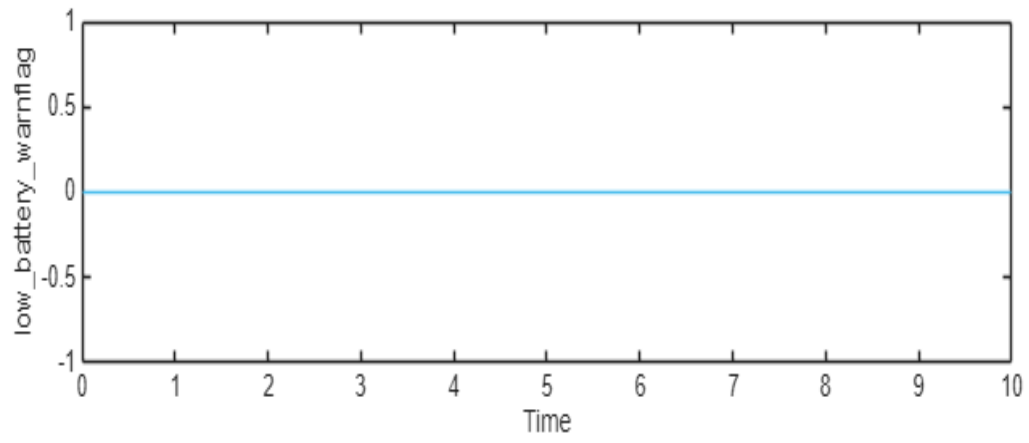
### System Under Test Information

|                           |                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Model:                    | SOC_Regler                                                                                                                                        |
| Release:                  | Current                                                                                                                                           |
| Simulation Mode:          | normal                                                                                                                                            |
| Override SIL or PIL Mode: | 0                                                                                                                                                 |
| Configuration Set:        | Configuration                                                                                                                                     |
| External Input Name:      | battery_current_input.mat                                                                                                                         |
| External Input File:      | C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04_Verification\input_data\battery_current_input.mat |
| Start Time:               | 0                                                                                                                                                 |
| Stop Time:                | 10                                                                                                                                                |
| Checksum:                 | 1580065241 1310189143 2029611893 136738514                                                                                                        |
| Simulink Version:         | 25.2                                                                                                                                              |
| Model Version:            | 1.4                                                                                                                                               |
| Model Author:             | 320236370                                                                                                                                         |
| Date:                     | Wed Apr 08 19:47:39 2026                                                                                                                          |
| User ID:                  | 320236370                                                                                                                                         |
| Model Path:               | C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC_Regler.slx                                       |
| Machine Name:             | YY309528                                                                                                                                          |
| Solver Name:              | ode45                                                                                                                                             |
| Solver Type:              | Variable-Step                                                                                                                                     |
| Max Step Size:            | 0.20000000000000001                                                                                                                               |
| Simulation Start Time:    | 2026-04-10 09:59:21                                                                                                                               |
| Simulation Stop Time:     | 2026-04-10 09:59:22                                                                                                                               |
| Platform:                 | PCWIN64                                                                                                                                           |

### Simulation Output

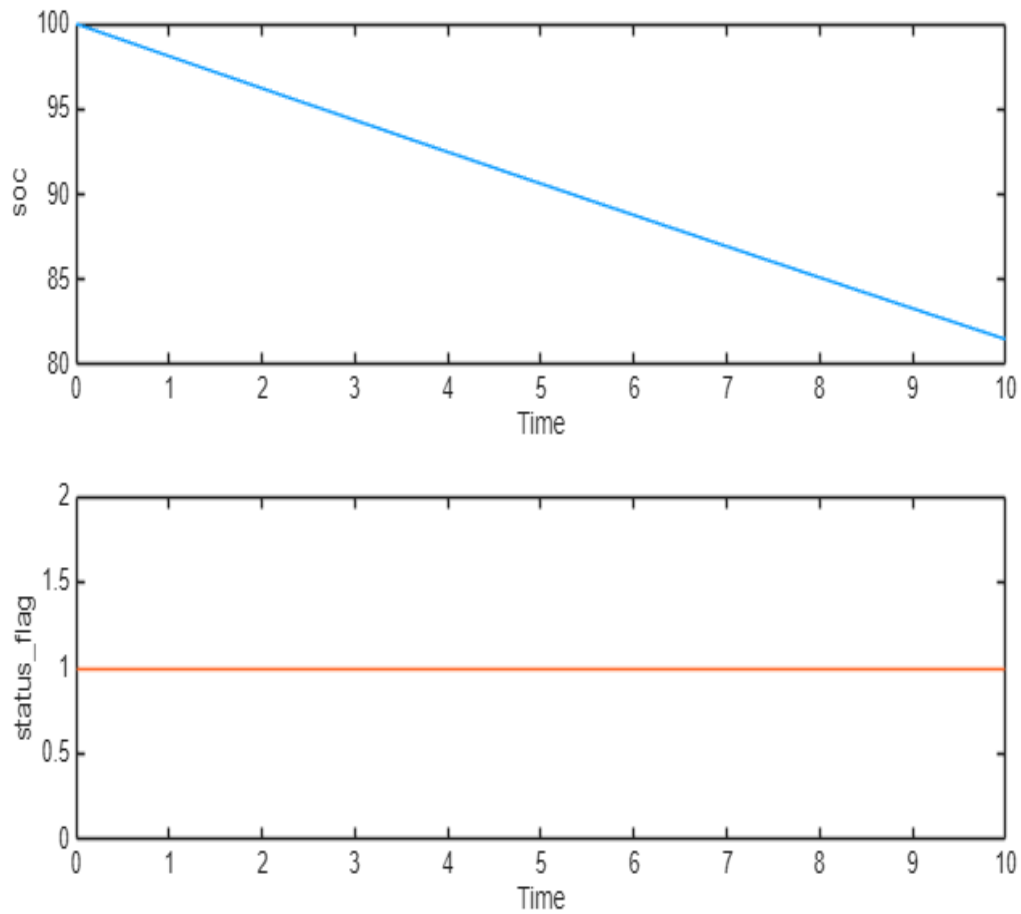
| Name                 | Data Type | Units | Sample Time | Interp | Sync  | Link to Plot         |
|----------------------|-----------|-------|-------------|--------|-------|----------------------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |
| soc                  | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| soc                  | double    |       | Continuous  | linear | union | <a href="#">Link</a> |
| status_flag          | boolean   |       | Continuous  | zoh    | union | <a href="#">Link</a> |

| Name                 | Data Type | Units | Sample Time | Interp | Sync  |
|----------------------|-----------|-------|-------------|--------|-------|
| low_battery_warnflag | boolean   |       | Continuous  | zoh    | union |
| soc                  | double    |       | Continuous  | linear | union |



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| Name        | Data Type | Units | Sample Time | Interp | Sync  |
|-------------|-----------|-------|-------------|--------|-------|
| soc         | double    |       | Continuous  | linear | union |
| status_flag | boolean   |       | Continuous  | zoh    | union |



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#### Simulation Logs:

Inputs may not be compatible for simulation. Test results might not be accurate. Click [here](#) for more information on external input mapping.

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